

PROJECT: Multi-AZ WordPress Deployment

Services: EC2, RDS (MySQL), EFS, ALB, Route 53, CloudFront

Goal: Highly available WordPress deployment with content delivery optimization.

Skills: Shared storage (EFS), DNS, caching strategies.

Definition: This project demonstrates the deployment of a highly available WordPress website on Amazon Web Services (AWS) using a multi-AZ architecture. The infrastructure integrates services such as Amazon EC2, Amazon RDS, Amazon EFS, Application Load Balancer (ALB), Route 53, and CloudFront to ensure scalability, reliability, and optimized content delivery. It showcases how cloud-native services can be combined to build a resilient and production-ready web application.

Scope of the Project: The project focuses on designing and implementing a fault-tolerant and scalable WordPress hosting environment in AWS. It covers configuring compute resources, managed databases, shared storage, DNS management, and global content delivery for improved performance. The implementation demonstrates real-world cloud architecture practices used to maintain high availability, load balancing, and efficient content distribution for web applications.

Methodologies:

1. Multi-AZ WordPress Deployment on AWS: Method used in this Project

- In this method, WordPress runs on Amazon EC2 instances with shared storage (EFS), while the database is managed using Amazon RDS (MySQL).
- Traffic is distributed using an Application Load Balancer, and Route 53 and CloudFront are used for DNS management and faster content delivery.
- This method is best because it provides high availability, scalability, and better performance. If one server or availability zone fails, the system can still continue running using other resources.
- Services like CloudFront and ALB also improve website speed and handle large amounts of traffic efficiently.

2. Serverless Deployment: Alternative Method

- In a serverless deployment, the application runs without managing servers by using services like AWS Lambda, API Gateway, and Amazon S3. The backend logic is executed through Lambda functions, while API Gateway handles incoming requests from users.
- Static website content such as HTML, CSS, and images can be hosted on Amazon S3. This method automatically scales based on traffic and reduces infrastructure management, but it may not be ideal for traditional applications like WordPress without additional configuration.

Limitations of Serverless Deployment

- Not ideal for traditional applications like WordPress because WordPress requires a persistent server environment and file system.
- Cold start delays can occur when Lambda functions are invoked after a period of inactivity.
- Execution time limits exist for Lambda functions, which may restrict long-running processes.
- More complex architecture and configuration compared to simple server-based deployments.

Services Used in this Project:

1. Amazon EC2 (Elastic Compute Cloud)

- **Definition:** Amazon EC2 is a cloud computing service that provides virtual servers to run applications on AWS.
- **Purpose:** EC2 is used to host and run the WordPress website with Apache and PHP.
- **Role in this Project:** EC2 acts as the web server that handles user requests and delivers the WordPress website.

2. Amazon RDS (Relational Database Service)

- **Definition:** Amazon RDS is a managed database service that allows users to easily create, manage, and maintain relational databases in the cloud.
- **Purpose:** Amazon RDS is used to store and manage the WordPress MySQL database.
- **Role in this Project:** Amazon RDS acts as the backend database that stores website data such as posts, users, and settings.

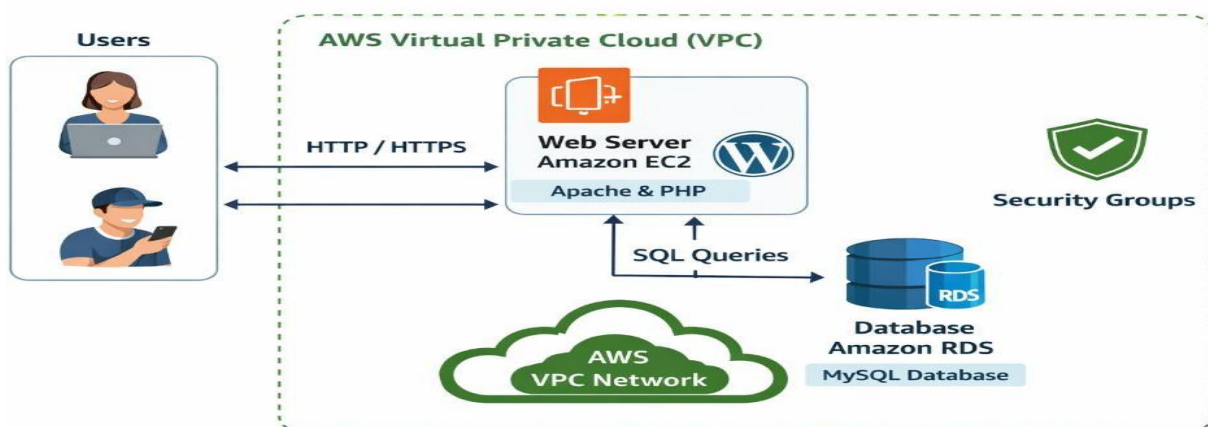
3. Security Group

- **Definition:** A Security Group is a virtual firewall in AWS that controls inbound and outbound network traffic for resources like EC2 and RDS.
- **Purpose:** The Security Group is used to allow necessary access such as HTTP, SSH, and database communication.
- **Role in this Project:** The Security Group ensures secure communication between the EC2 instance and the RDS database while protecting the server from unauthorized access.

4. VPC (Virtual Private Cloud)

- **Definition:** A VPC (Virtual Private Cloud) is a virtual network in AWS where cloud resources like EC2 and RDS are securely deployed.
- **Purpose:** The VPC provides the network environment for hosting the EC2 instance and RDS database.
- **Role in this Project:** The VPC enables secure communication between EC2 and RDS within the same network.

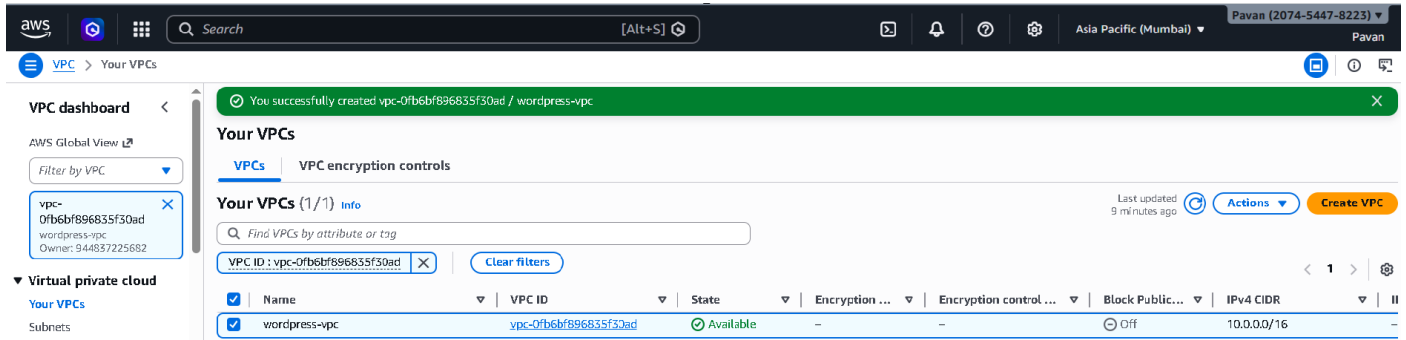
Architecture Diagram:



Implementation and Execution of Multi-AZ WordPress Deployment.

STEP 1: Create VPC

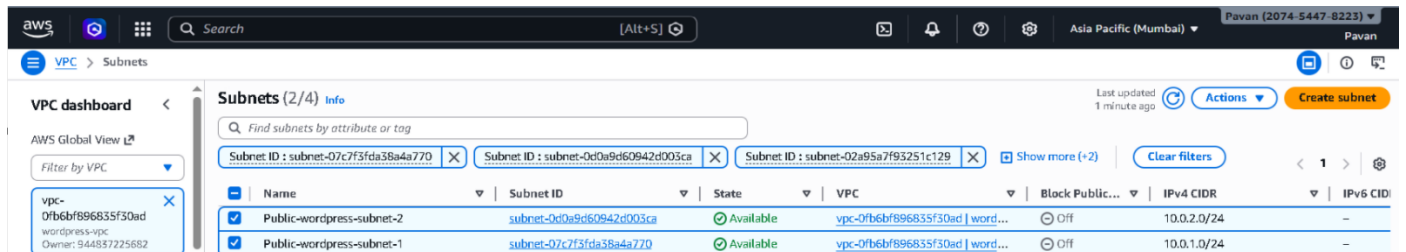
- Name: wordpress-vpc
- IPV4 CIDR: 10.0.0.0/16
- Click create vpc



STEP 2: Create Subnets

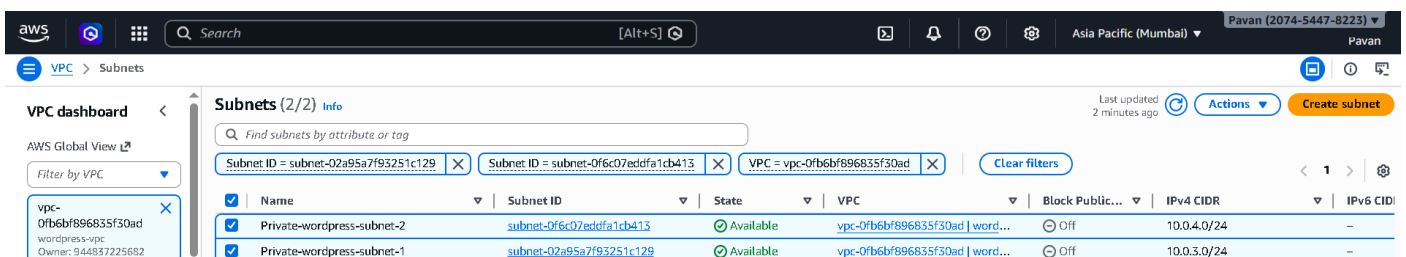
Public Subnets

- Select vpc: wordpress-vpc
- Subnet name: public-wordpress-subnet-1 and public-wordpress-subnet-2
- Availability Zone: ap-south-1a and ap-south-1b
- IPV4 CIDR: 10.0.1.0/24 and 10.0.2.0/24
- Click create subnet



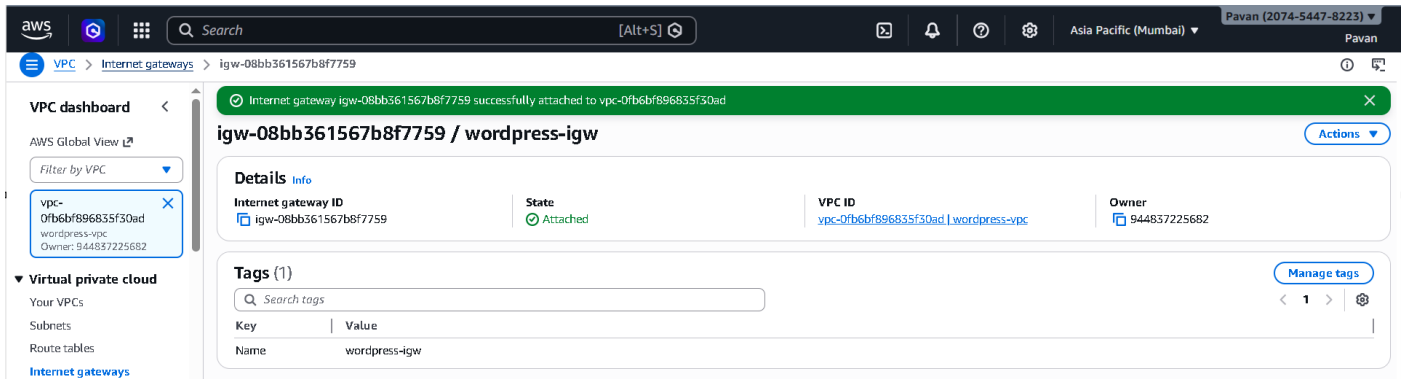
STEP 3: Private Subnets

- Select vpc: wordpress-vpc
- Subnet name: private-wordpress-subnet-1 and private-wordpress-subnet-2
- Availability Zone: ap-south-1a and ap-south-1b
- IPV4 CIDR: 10.0.3.0/24 and 10.0.4.0/24
- Click create subnet



STEP 4: Create Internet Gateway

- Name: wordpress-igw
- Click Create Internet Gateway
- Select IGW: Actions: Attach to VPC
- Click on Attach

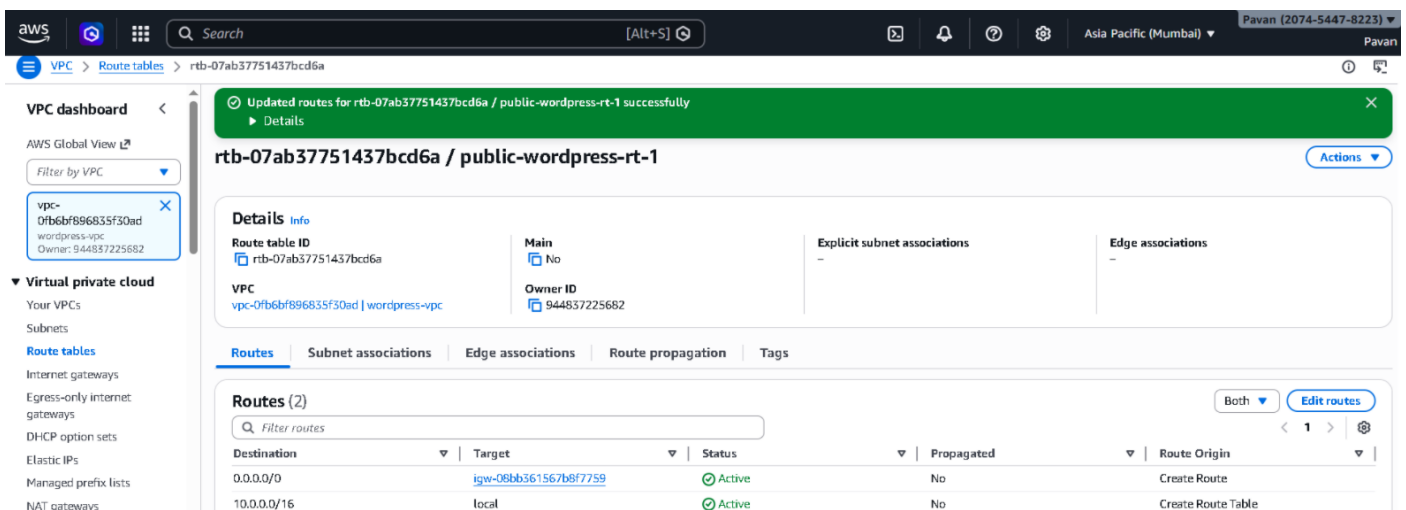


STEP 5: Create Public Route Table

- Name: public-wordpress-rt-1
- VPC: wordpress-vpc
- Click create route table

Add Route:

- Select public-wordpress-rt-1
- Routes: Edit routes
- Add routes:
 - Destination: 0.0.0.0/0
 - Target: Internet Gateway(wordpress-igw)
 - Save routes



Associate Public Subnet:

- Subnet Associations: Edit
- Select public-wordpress-subnet 1 and public-wordpress-subnet-2
- Save

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VPC > Route tables > rtb-07ab37751437bcd6a

VPC dashboard

AWS Global View

Filter by VPC

vpc-0fb6bf896835f30ad
wordpress-vpc
Owner: 944837225682

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

You have successfully updated subnet associations for rtb-07ab37751437bcd6a / public-wordpress-rt-1.

rtb-07ab37751437bcd6a / public-wordpress-rt-1 Actions

Details Info

Route table ID: rtb-07ab37751437bcd6a

Main: No

Explicit subnet associations: 2 subnets

Edge associations: -

VPC: vpc-0fb6bf896835f30ad | wordpress-vpc

Owner ID: 944837225682

Routes Subnet associations Edge associations Route propagation Tags

Explicit subnet associations (2) Edit subnet associations

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Public-wordpress-subnet-2	subnet-0d0a9d60942cd003ca	10.0.2.0/24	-
Public-wordpress-subnet-1	subnet-07c7f3fda38a4a770	10.0.1.0/24	-

STEP 6: Created Subnet Group

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Aurora and RDS > Subnet groups

Aurora and RDS

Dashboard

Databases

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Successfully created wordpress-db-subnet-group. View subnet group

Subnet groups (1) Edit Delete Create DB subnet group

Filter by subnet group

Name	Description	Status	VPC
wordpress-db-subnet-group	Subnet group for wordpress RDS database	Complete	vpc-0fb6bf896835f30ad

STEP 7: Created Database

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Aurora and RDS > Databases

Aurora and RDS

Dashboard

Databases

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Successfully created database wordpress-db. View connection details

You can use settings from wordpress-db to simplify configuration of suggested database add-ons while we finish creating your DB for you.

Databases (1) Group resources Modify Actions Create database

Filter by databases

DB identifier	Status	Role	Engine	Upgrade rollout order	Region...	Size	Recommendation
wordpress-db	Available	Instance	MySQL Co...	SECOND	ap-south-1b	db.t3.micro	

STEP 8: Created Elastic File System

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Amazon EFS > File systems

Elastic File System

File systems

Access points

AWS Backup

AWS DataSync

AWS Transfer

Documentation

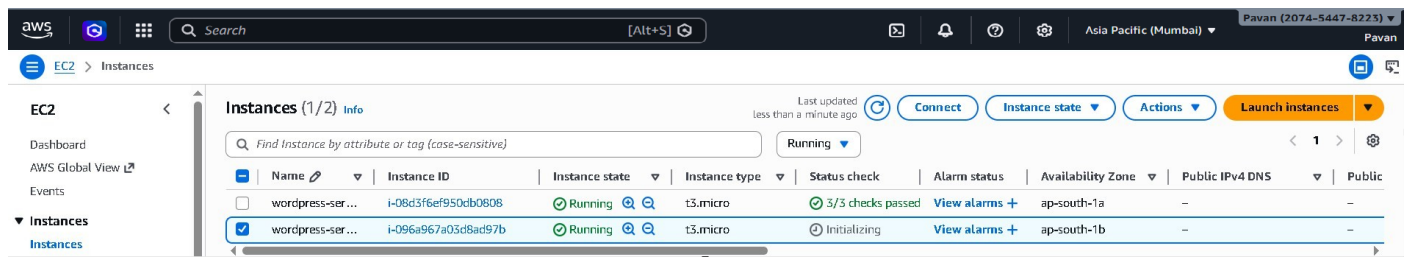
Success! File system (fs-05e7dcbad1a676b4c) is available. View file system

File systems (1) View details Delete Create file system

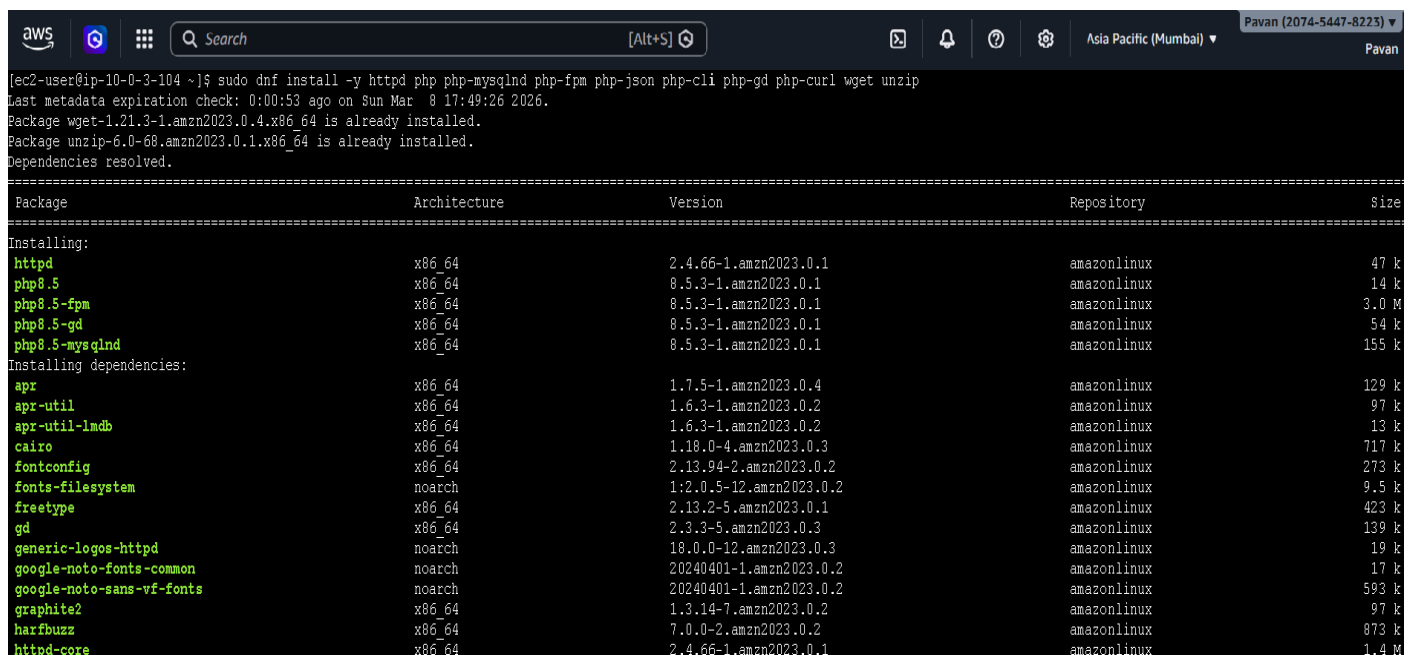
Filter by property values

Name	File system ID	Encrypte d	Total size	Size in Standard	Size in IA	Size in Archive	Provisioned Throughput (MiB/s)	File system state	Creation time
wordpress-efs	fs-05e7dcbad1a676b4c	Encrypte d	6.00 KiB	6.00 KiB	0 Bytes	0 Bytes	-	Available	Sun, 08 Mar 2026 16:34:10 GMT

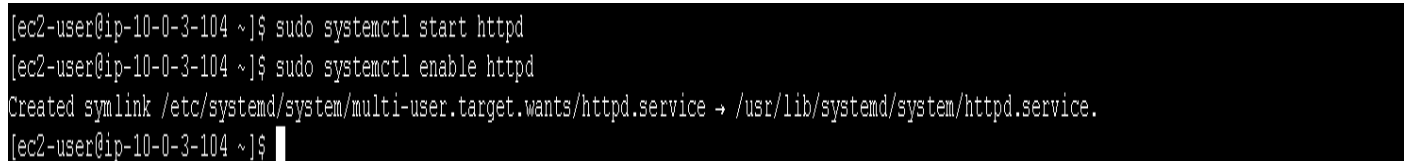
STEP 9: Launch an instance



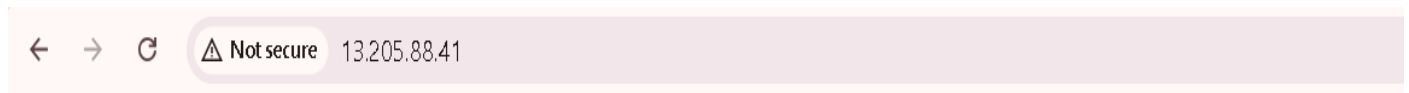
STEP 10: Update the server and Install Apache + PHP + MySQL client



STEP 11: Start Apache and enable



STEP 12: Test Apache



It works!

STEP 13: Download wordpress

```
[ec2-user@ip-10-0-3-104 html]$ sudo wget https://wordpress.org/latest.tar.gz
--2026-03-08 17:55:45-- https://wordpress.org/latest.tar.gz
Resolving wordpress.org (wordpress.org)... 198.143.164.252, 2607:f978:5:8002::c68f:a4fc
Connecting to wordpress.org (wordpress.org)|198.143.164.252|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 27062929 (26M) [application/octet-stream]
Saving to: 'latest.tar.gz'

latest.tar.gz                               100%[=====>] 25.81M  8.05MB/s   in 3.9s

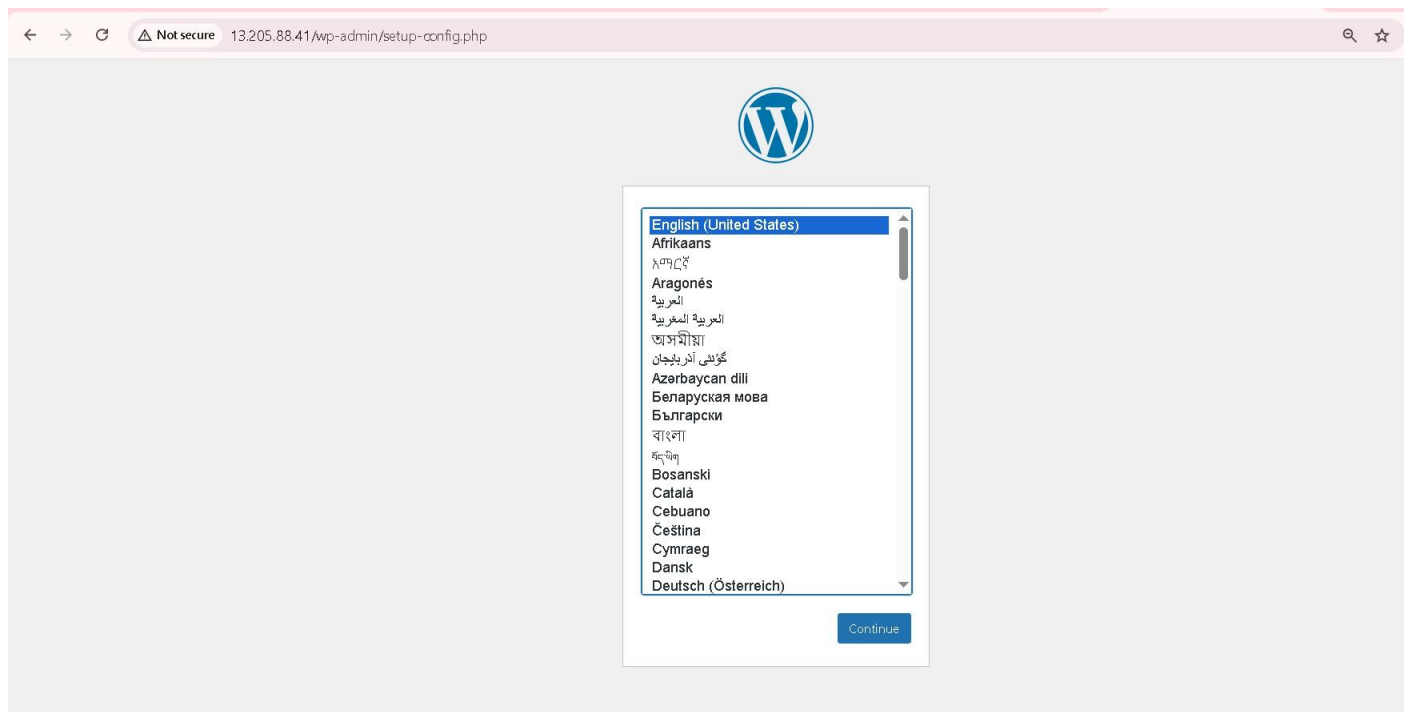
2026-03-08 17:55:50 (6.63 MB/s) - 'latest.tar.gz' saved [27062929/27062929]

[ec2-user@ip-10-0-3-104 html]$
```

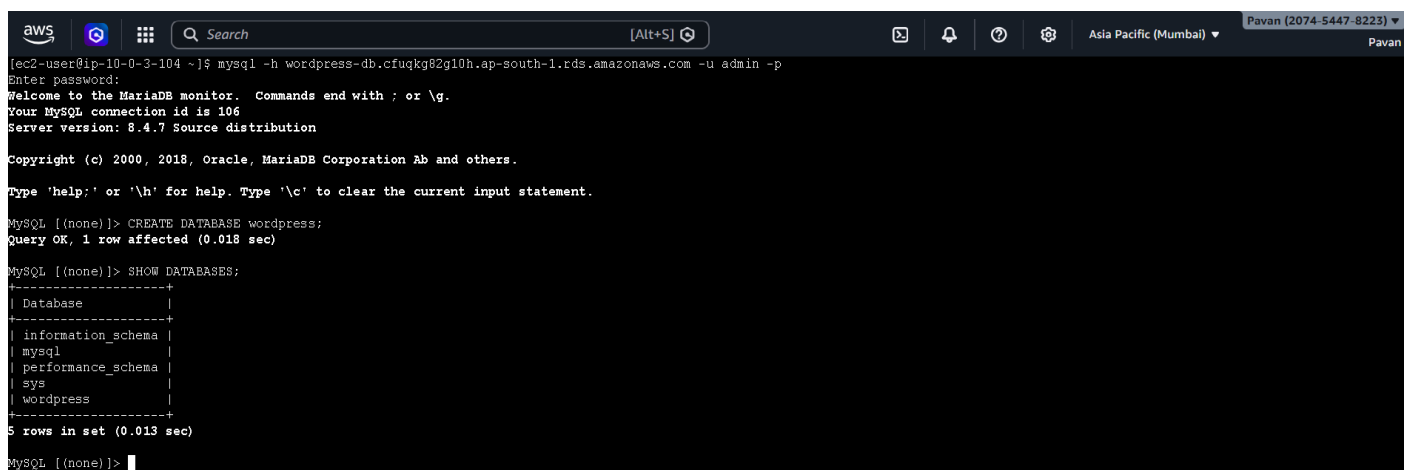
STEP 14: Restart Apache

```
[ec2-user@ip-10-0-3-104 html]$ sudo systemctl restart httpd
[ec2-user@ip-10-0-3-104 html]$
```

STEP 15: We can see wordpress installation page



STEP 16: We will connect wordpress-→ RDS database



← → ↻ ⚠ Not secure 13.205.88.41/wp-admin/setup-config.php?step=1



Below you should enter your database connection details. If you are not sure about these, contact your host.

Database Name	wordpress
The name of the database you want to use with WordPress.	
Username	admin
Your database username.	
Password	***** Show
Your database password.	
Database Host	wordpress-db.cfugkg82eg10h.ap-south-1.rds.amazonaws.com
You should be able to get this info from your web host, if localhost does not work.	
Table Prefix	wp_
If you want to run multiple WordPress installations in a single database, change this.	

[Submit](#)

← → ↻ ⚠ Not secure 13.205.88.41/wp-admin/install.php?step=2



Success!

WordPress has been installed. Thank you, and enjoy!

Username	admin
Password	Your chosen password.

[Log In](#)

← → ↻ ⚠ Not secure 13.205.88.41/wp-admin/ 🔍 ☆ 📁 ⓘ Ⓢ ⋮

 My AWS WordPress Site 0 + New

Howdy, admin

Dashboard

Home

Updates

Posts

Media

Pages

Comments

Appearance

Plugins

Users

Tools

Settings

Collapse Menu

Site Health Status

No information yet...

Site health checks will automatically run periodically to gather information about your site. You can also [visit the Site Health screen](#) to gather information about your site now.

At a Glance

1 Post 1 Page

1 Comment

WordPress 6.9.1 running [Twenty Twenty-Five](#) theme.

Activity

Recently Published

Today, 6:58 pm Hello world!

Recent Comments

From A WordPress Commenter on Hello world! Hi, this is a comment. To get started with moderating, editing, and deleting comments, please visit the Comments screen in...

All (1) | Mine (0) | Pending (0) | Approved (1) | Spam (0) | Trash (0)

Quick Draft

Title

Content

What's on your mind?

Save Draft

WordPress Events and News

Attend an upcoming event near you. [Select location](#)

There are no events scheduled near you at the moment. Would you like to [organize a WordPress event?](#)

WordPress 7.0 Beta 3

WordPress 7.0 Beta 2

Gutenberg Times: PHP-only blocks, WordCamp Asia, Dev Notes for WordPress 7.0 — Weekend Edition #360

Matt: Declaration of the Independence

Open Channels FM: Open Channels FM v7.1 Release

Meetups WordCamps News

Drag boxes here

Thank you for creating with [WordPress](#).

Version 6.9.1

Troubleshooting:

1. WordPress Database Connection Error

- **Issue:** WordPress showed “Error establishing database connection” during setup.
- **Solution:** Updated the wp-config.php file with the correct database details:

- DB Name
- DB Username
- DB Password
- RDS Endpoint

2. Unable to Connect EC2 to RDS Database

- **Issue:**
The EC2 instance could not connect to the MySQL database hosted on Amazon RDS.
- **Solution:**
Added a MySQL rule (Port 3306) in the RDS security group and allowed access from the EC2 security group.

Conclusion:

This project successfully demonstrated the deployment of a WordPress website using Amazon EC2 and Amazon RDS on AWS cloud infrastructure. The implementation showed how cloud services can be used to build a scalable, reliable, and cost-effective web hosting environment. Through this project, we gained practical experience in cloud computing, server configuration, database connectivity, and security management. The architecture allows the website to be easily managed and accessed publicly through the internet. Overall, the project highlights how AWS services can simplify the process of deploying and managing modern web applications.